

Text Image GenAI methods flashcards

STUDY PACK

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Format: question and answer study cards

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Q. What are the main components of the Text-Image GenAI pipeline?

A. Text encoder, latent diffusion model (text→image), vision-language captioner (image→text), evaluation/API.

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Q. What advantage do latent diffusion models provide?

A. Operate in compressed latent space for high fidelity sampling and computational efficiency compared to pixel-space diffusion.

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Q. What is classifier-free guidance?

A. A technique that mixes conditional and unconditional denoising to trade fidelity vs diversity via a guidance scale.

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Q. What metric measures semantic alignment of image and text?

A. CLIPScore.

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Q. How is round-trip evaluation performed?

A. Generate image from prompt → caption the image → compare caption to original prompt (semantic metrics).

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Q. Name the multi-objective loss components used.

A. L_{text} (CLIP contrastive), L_{img} (diffusion reconstruction), L_{perc} (perceptual), $R(\Theta)$ (regularization).

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Q. Which metric evaluates image quality?

A. FID (Fréchet Inception Distance).

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Q. Two engineering strategies to reduce compute cost?

A. Use latent space sampling, mixed precision, caching embeddings, and smaller checkpoints or adapters.

**Q. Name one
safety control to
add in
deployment.**

A. Content filters,
watermarking, and
human review.